Lesson 15: Experimentation and Validity

English Stage 4 (Psychology) · Unit 3: Research Designs

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 5.4: Experimentation and Validity.

In this lesson

You will learn the "four big validities" used to judge whether an experiment is sound: internal, external, construct and statistical validity. You will evaluate famous studies with them, see why researchers must prioritise, and practise hedging, the language that makes academic claims appropriately careful.

Learning goals

After this lesson you should be able to:

- Explain internal validity and why experiments are high in it.
- Evaluate a study's external validity, using mundane and psychological realism.
- Explain construct validity and statistical validity, with an example of each.
- Explain why researchers must prioritise some validities over others.
- Use hedges to express evaluations and conclusions with appropriate caution.

Key language

Term or phrase	Meaning	Example
internal validity	the extent to which the design supports the conclusion that the IV caused differences in the DV	Experiments are high in internal validity.
external validity	the extent to which results can be generalised to people and situations beyond those studied	Would the result hold for other guests in other hotels?
mundane realism	similarity of the participants and situation to those people encounter every day	real shoppers in a real grocery store
psychological realism	the same mental process is used in the laboratory and in the real world	the visual processing of colours
construct validity	the quality of the experiment's manipulations: how well they represent the research question	Increasing the number of potential helpers tested diffusion of responsibility.

Term or phrase	Meaning	Example
operationalization	turning a research question into an experimental design (see operational definition)	Darley and Latané operationalized diffusion of responsibility.
statistical validity	proper statistical treatment of the data and sound statistical conclusions	the right test, its assumptions met, enough participants
power analysis	a calculation of the number of participants needed to detect an effect of a specific size	It should be done when designing a study.
field experiment	an experiment conducted outside the laboratory	Cialdini's hotel towel study
correlation does not imply causation	two related variables are not necessarily cause and effect	Happier people may exercise more.
it seems (very) likely that	a hedge: the claim is probable, not certain	It seems very likely that their result would hold true for other guests.
draw the blanket conclusion that	make an over-general judgement	We should be careful not to draw the blanket conclusion that experiments are low in external validity.

Rules and explanations

The ideas

Four big validities

Researchers (Judd & Kenny, 1981; Morling, 2014) assess experiments on four validities:

Validity	Question it asks
Internal	Does the way the study was conducted show that the IV caused the change in the DV?
External	Can the results be generalised to other people and situations?
Construct	Do the manipulations represent the research question well?
Statistical	Were the right statistics used, and do they support the conclusions?

Internal validity

Two related variables do not necessarily have a causal relationship. If people who exercise are happier, perhaps happiness causes exercise, or better physical health causes both. An experiment creates highly similar conditions and changes only the IV, so a later difference must have been caused by the IV (in Darley and Latané's study, the number of students that participants believed were present).

Manipulation and control (for example, random assignment) give experiments **high internal validity**; non-experimental designs such as correlational studies are low in it.

External validity

Manipulation and control can make experiments seem artificial: undergraduates in a laboratory filling in questionnaires. In Fredrickson et al. (1998), students took a math test while wearing a swimsuit. Studies are higher in external validity when the participants and situation resemble those the researchers want to generalise to and those people meet every day (**mundane realism**). A cereal-box colour study with real shoppers in a real store has high external validity; a laboratory study of students judging colours on a screen has low external validity but high **psychological realism**, because colour processing is the same in both places.

Two reasons not to conclude that experiments are low in external validity:

1. **They need not seem artificial.** Darley and Latané simulated a real emergency quite well. In a **field experiment**, Cialdini (2005) varied the message on a card in hotel rooms: respect for the environment, a donation by the hotel, or "most hotel guests reuse their towels". Guests given the last message reused their towels substantially more often.
2. **They often study general psychological processes.** In Fredrickson's study women, but not men, did worse in swimsuits. The researchers explained this by self-objectification (thinking about oneself from an outside observer's perspective), which takes attention away from other tasks and probably operates in many women and situations.

Construct validity

Construct validity concerns the quality of the manipulations. Darley and Latané's research question ("does helping behavior become diffused?") was **operationalized** by increasing the number of potential helpers; with a crisis, a way to help and more students in the discussion, construct validity was very high. With only two conditions (one or two other students), a decrease in helping might show only the presence of others, so construct validity would be lower. More conditions might reveal a continuing decrease or a plateau, but they do not automatically raise construct validity.

Statistical validity

Statistical validity concerns the right test (*t* test, ANOVA, regression, correlation) for the measurement scale of the DV and the design, and whether the test's assumptions (for example, normally distributed data) are met. The common criticism "too few participants" is a statistical-validity criticism, not an external-validity one, because small samples (even of one person) can be illuminating. The chance of detecting an effect depends on the number of conditions and the sample size, so researchers should run a **power analysis** when designing a study.

Prioritizing validities

It is often impossible to be high on all four. Cialdini's study was high in external validity but more modest in statistical validity; this does not invalidate it but shows where follow-up studies can improve. Morling (2014) notes that many studies are high in internal and construct validity but sometimes sacrifice external validity.

Language focus

Hedging in evaluation. The section judges studies carefully, so it uses hedges: words that show how certain a claim is.

- Modal verbs: "Even though we **may** see a decrease in helping ..., it **may not** be a clear demonstration of diffusion of responsibility."
- Likelihood expressions: "it **seems very likely that** their result would hold true for other guests in other hotels"; "this process ... **is likely to** operate in a variety of women and situations".
- Adverbs: "**perhaps** it would plateau after a certain number of people".
- Confidence: "the researchers **would be fairly confident** that this increase would be true for other shoppers".

Hedges do not make writing weak; they make claims match the evidence. Notice also that negative evaluation is balanced: "This discrepancy **does not invalidate** the study **but** it shows where there may be room for improvement."

> **Note for Arabic speakers** - Formal Arabic often strengthens claims with emphatic words, and English readers may read a direct translation as overstatement. In English academic writing, check every strong verb (*proves*, *shows clearly*) and add a hedge where the evidence is limited.

Practice

1. Reading questions

1. Why are experiments high in internal validity, and why are correlational designs low?
2. If people who exercise are happier, what two alternative explanations does the book give?
3. In the cereal-box example, which study is higher in external validity? What does the other study still have?
4. In Cialdini's hotel study, what was manipulated, and what was the result?
5. What did Fredrickson and colleagues find, and how did they explain it?
6. Why would a two-condition version of Darley and Latané's experiment have lower construct validity?
7. Why does the book treat "not enough participants" as a statistical-validity problem? What does a power analysis tell you?
8. According to Morling (2014), which validities do many psychology studies have, and which do they sometimes sacrifice?

2. Apply the four validities

This section has no exercises of its own (the chapter exercises are in Section 5.6), so these tasks use the section's questions and studies.

1. The book asks: "When will undergraduate students ever have to complete math tests in their swimsuits outside of this experiment?" Explain why this criticism does not make the study useless.
2. Rate Darley and Latané's experiment on each of the four validities, using Lessons 13 and 15. Where the text gives no evidence, say so.
3. Cialdini's statistical validity was "more modest". Suggest two ways a follow-up study could improve it.

3. Language exercise

A. Find the hedges in these sentences from the section.

1. "They argued, furthermore, that this process of self-objectification and its effect on attention is likely to operate in a variety of women and situations."
2. "... the researchers would not be very confident that this preference is relevant to grocery shoppers' cereal-buying decisions ..."
3. "In that situation, we may develop a more nuanced understanding of the phenomenon."

B. Rewrite these overconfident claims with suitable hedges.

4. Experiments are low in external validity.
5. The towel message will work in every hotel.
6. Small samples make results wrong.

Speak with your tutor

1. Explain the four validities to your tutor using only Cialdini's hotel study, then ask which explanation was least clear.
2. In PSYC 110 Lecture 9, Professor Salovey calls the "rickety bridge" field study "bad science" because there was no random assignment, and then describes laboratory versions with random assignment. Discuss the internal and external validity of the two kinds of study.
3. Defend Fredrickson's swimsuit study against a tutor who says it is "too artificial to matter". Use the terms *psychological realism* and *mundane realism*.
4. Tell your tutor which validity you will prioritise in your sleep and memory study, and why. Ask the tutor to challenge you.

Writing task

Research project: the validity of your design (Lessons 13-16 build your design and its validity).

Write 200-250 words evaluating the experiment you planned in Lessons 13 and 14. Give two or three sentences on each validity:

- **internal:** how manipulation, random assignment or counterbalancing, and control support a causal conclusion;
- **external:** who and what your results can be generalised to (university students only? a nap in a laboratory?);
- **construct:** how well your manipulation of sleep represents your research question;
- **statistical:** how you will decide the sample size (a power analysis: are 20 students enough?) and choose a test that fits your DV and design.

End with the validity you prioritise and the one you partly sacrifice, and why.

Language: hedges (*may*, *is likely to*, *it seems likely that*), evaluative phrases (*is high in*, *at the expense of*, *does not invalidate … but*).

If you work on your own topic, evaluate your own design in the same way.

Self-editing checklist:

- All four validities are discussed, each with a specific feature of your design.
- Small sample size is treated as a statistical-validity issue.
- The trade-off between internal and external validity is stated.
- Claims about generalisation are hedged.
- No claim says the study will "prove" anything.
- *Valid* (adjective) and *validity* (noun) are used correctly.

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- Morling (2014), the APS Observer article behind the "four big validities" (the book's link is broken; this is the current address): https://www.psychologicalscience.org/observer/teach-your-students-to-be-better-consumers
- Cialdini (2005), "Don't Throw in the Towel", on the hotel field experiment (current address): https://www.psychologicalscience.org/observer/dont-throw-in-the-towel-use-social-influence-research
- PSYC 110 Lecture 9 (the bridge study and its laboratory versions): https://oyc.yale.edu/psychology/psyc-110/lecture-9

Answer key

1. Reading questions

1. Experiments manipulate the IV and control extraneous variables (for example by random assignment), which strongly supports causal conclusions; in correlational designs variables are measured, not manipulated ("Internal Validity", paragraph 3).
2. Greater happiness may cause people to exercise, or something like better physical health may cause both exercise and happiness ("Internal Validity", paragraph 1).
3. Studying ordinary people doing their weekly shopping in a real grocery store is higher in external validity and mundane realism. The laboratory colour study still has high psychological realism ("External Validity", paragraph 2).
4. The message on a card in hotel rooms (environment, donation, most guests reuse towels). Guests told that most guests reuse their towels reused their own substantially more often ("External Validity", paragraph 3).
5. Women, but not men, performed worse on the math test in swimsuits. They explained it by self-objectification, which diverts attention from other tasks ("External Validity", paragraph 4).
6. A decrease from one to two others might show only the effect of the presence of others (perhaps social inhibition), not a clear demonstration of diffusion of responsibility ("Construct Validity", paragraph 2).
7. Some studies with very small samples, even of one person, are illuminating, so small samples are not in themselves an external-validity problem. A power analysis tells you how many participants you need to detect an effect of a specific size ("Statistical Validity").
8. High internal and construct validity; they sometimes sacrifice external validity ("Prioritizing Validities").

2. Apply the four validities

1. (example answer) The swimsuit is not meant to copy daily life. It is a way to cause self-objectification, a process that probably affects many women in many situations, so the result can generalise even though the situation never occurs outside the laboratory.
2. (example answer) Internal: high, because the number of students participants believed were present was the only difference between conditions. External: fairly good, because it was a reasonably good simulation of a real emergency, although the participants were all university students. Construct: very high, according to the book, because the manipulation spoke clearly to diffusion of responsibility. Statistical: the section gives no information about the analysis or the sample size, so it cannot be judged from this text.
3. (example answer) Test more hotels and rooms after a power analysis to choose the sample size; choose a statistical test that suits the scale of the DV (whether or not a towel was reused) and check its assumptions.

3. Language exercise

1. *argued*, *is likely to*. 2. *would not be very confident*. 3. *may*.
4. (example answer) Experiments may be lower in external validity when their conditions are artificial, but this is not always the case.
5. (example answer) The social-norm message is likely to increase towel reuse in many other hotels.
6. (example answer) Small samples may reduce statistical validity, because it is difficult to generalise about a population from a small sample.

My notes